

ĐẠI HỌC BÁCH KHOA HÀ NỘI
TRƯỜNG CÔNG NGHỆ THÔNG TIN VÀ TRUYỀN
THÔNG

BÁO CÁO THỰC HÀNH LAB 01
IT1130-744360-2024.1



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MỤC LỤC

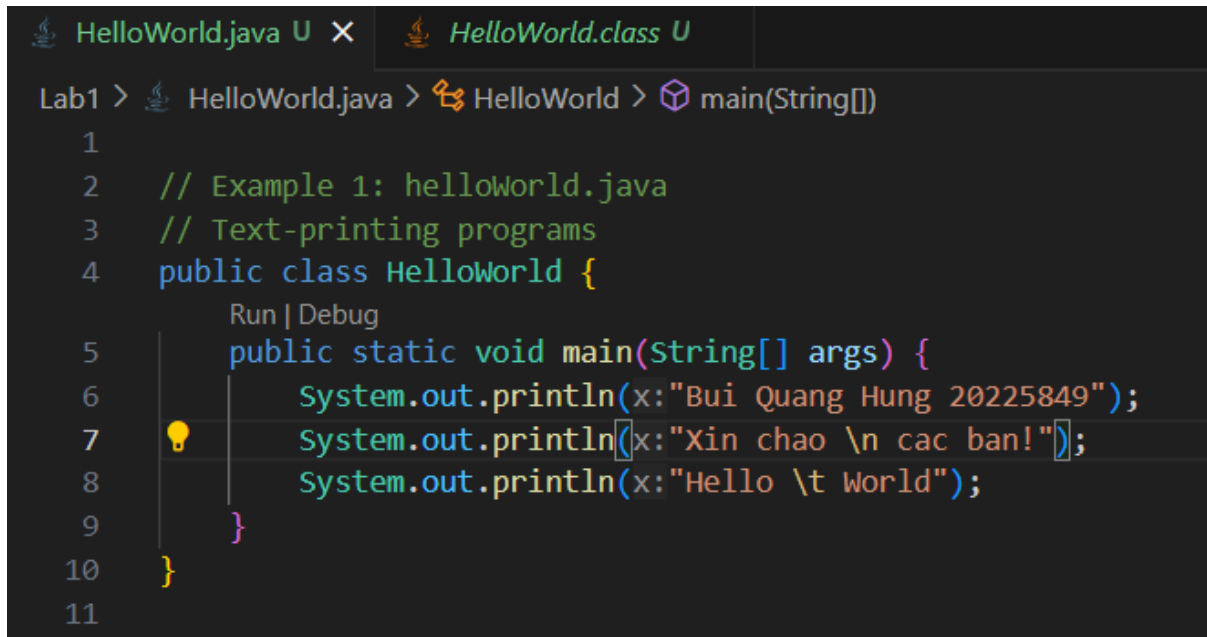
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A. The Very First Java Programs

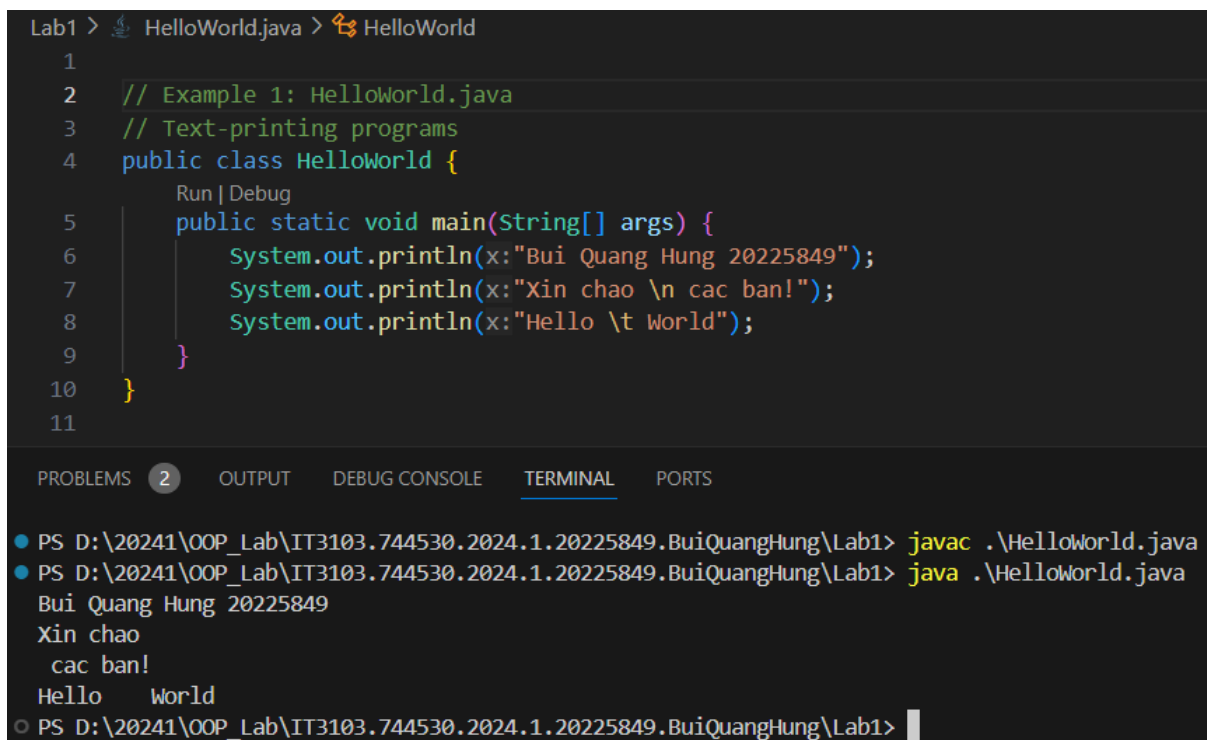
2.2.1 Write, compile the first Java application:

a. Code



```
Lab1 > HelloWorld.java > HelloWorld > main(String[])
1
2 // Example 1: HelloWorld.java
3 // Text-printing programs
4 public class HelloWorld {
5     public static void main(String[] args) {
6         System.out.println(x:"Bui Quang Hung 20225849");
7         System.out.println(x:"Xin chao \n cac ban!");
8         System.out.println(x:"Hello \t World");
9     }
10 }
11
```

b. Kết quả



```
Lab1 > HelloWorld.java > HelloWorld
1
2 // Example 1: HelloWorld.java
3 // Text-printing programs
4 public class HelloWorld {
5     public static void main(String[] args) {
6         System.out.println(x:"Bui Quang Hung 20225849");
7         System.out.println(x:"Xin chao \n cac ban!");
8         System.out.println(x:"Hello \t World");
9     }
10 }
11
```

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
● PS D:\20241\OOP_Lab\IT3103.744530.2024.1.20225849.BuiQuangHung\Lab1> javac .\HelloWorld.java
● PS D:\20241\OOP_Lab\IT3103.744530.2024.1.20225849.BuiQuangHung\Lab1> java .\HelloWorld.java
Bui Quang Hung 20225849
Xin chao
cac ban!
Hello    World
○ PS D:\20241\OOP_Lab\IT3103.744530.2024.1.20225849.BuiQuangHung\Lab1>
```

2.2.2 Write, compile the first dialog Java program

a. Code

```
FirstDialog.java U X
Lab1 > FirstDialog.java > FirstDialog > main(String[])
3 import javax.swing.JOptionPane;
4 public class FirstDialog {
5     public static void main(String[] args) {
6         JOptionPane.showMessageDialog(parentComponent:null, "Bui Quang Hung 20225849 "
7             + "Hello World! How are you?");
8         System.exit(status:0);
9     }
10 }
11
```

b. Kết quả

```
Lab1 > FirstDialog.java > FirstDialog > main(String[])
1
2 // Example2: fisrtDialog.java
3 import javax.swing.JOptionPane;
4 public class FirstDialog {
5     public static void main(String[] args) {
6         JOptionPane.showMessageDialog(parentComponent:null, "Bui Quang Hung 20225849 "
7             + "Hello World! How are you?");
8     }
9 }
10
```

PROBLEMS 2 OUTPUT

PS D:\20241\OOP_Lab\IT3103.744530.2024.1.20225849.BuiQuangHung\Lab1> java .\FirstDialog.java

Message

Bui Quang Hung 20225849 Hello World! How are you?

OK

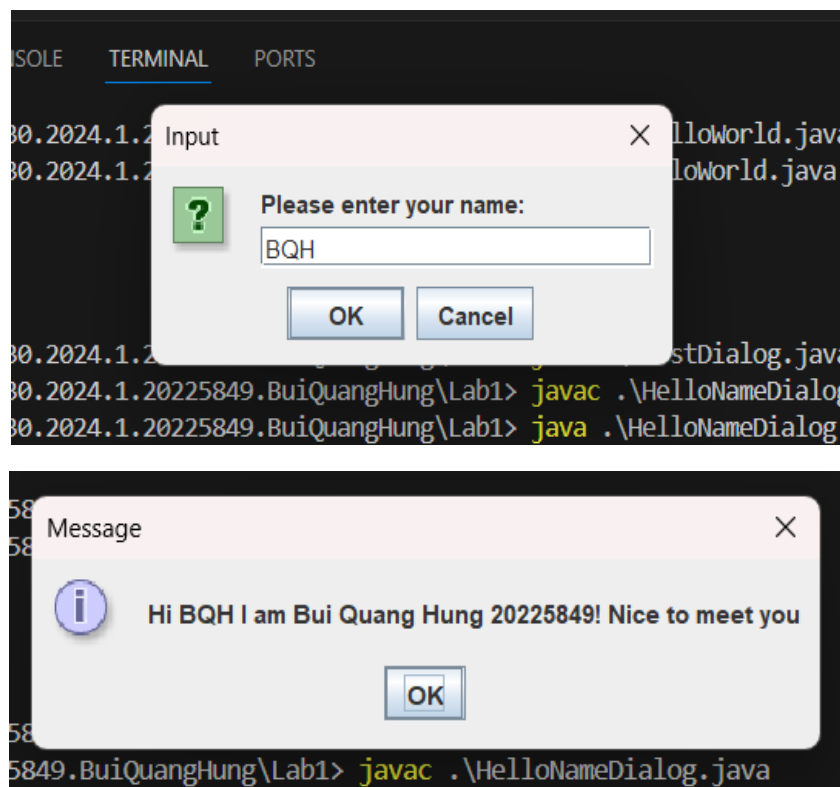
.\HelloWorld.java
HelloWorld.java

2.2.3 Write, compile the first input dialog Java application

a. Code

```
FirstDialog.java U HelloNameDialog.java U X
Lab1 > HelloNameDialog.java > HelloNameDialog > main(String[])
1 // Example 3: HelloNameDialog.java
2 import javax.swing.JOptionPane;
3 public class HelloNameDialog {
4     public static void main(String[] args) {
5         String result;
6         result = JOptionPane.showInputDialog(message:"Please enter your name:");
7         JOptionPane.showMessageDialog(parentComponent:null, "Hi "
8             + result + " I am Bui Quang Hung 20225849! Nice to meet you");
9         System.exit(status:0);
10    }
11 }
```

b. Kết quả

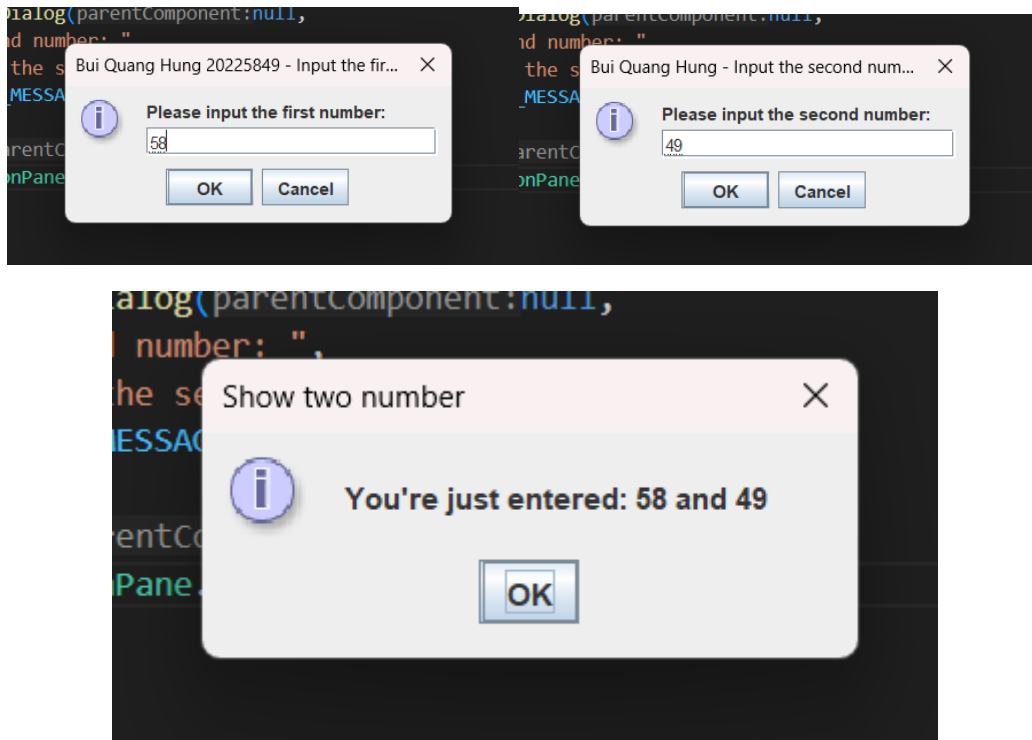


2.2.4 Write, compile, and run the following example:

a. Code

```
Lab1 > ShowTwoNumber.java U X
ShowTwoNumber > main(String[])
1
2 // Example 4: ShowTwoNumber.java
3 import javax.swing.JOptionPane;
4
5 public class ShowTwoNumber {
6     public static void main(String[] args) {
7         String strNum1, strNum2;
8         String strNotification = "You're just entered: ";
9         strNum1 = JOptionPane.showInputDialog(parentComponent:null,
10             message:"Please input the first number: ",
11             title:"Bui Quang Hung 20225849 - Input the first number", JOptionPane.INFORMATION_MESSAGE);
12         strNotification += strNum1 + " and ";
13         strNum2 = JOptionPane.showInputDialog(parentComponent:null,
14             message:"Please input the second number: ",
15             title:"Bui Quang Hung - Input the second number",
16             JOptionPane.INFORMATION_MESSAGE);
17         strNotification += strNum2;
18         JOptionPane.showMessageDialog(parentComponent:null, strNotification,
19             title:"Show two number", JOptionPane.INFORMATION_MESSAGE);
20         System.exit(status:0);
21     }
22 }
```

b. Kết quả



2.2.5 Write a program to calculate sum, difference, product, and quotient of 2 double numbers which are entered by users.

a. Code

```
Lab1 > Calculate.java > ...
1 // Example 5: Calculate.java
2 import java.util.Scanner;
3 public class Calculate {
4     Run | Debug
5     public static void main(String[] args){
6         Scanner sc = new Scanner(System.in);
7         System.out.print(s:"Bui Quang Hung 20225849 - Please enter first number: ");
8         double firstNum = sc.nextDouble();
9         System.out.print(s:"Bui Quang Hung 20225849 - Please enter second number: ");
10        double secondNum = sc.nextDouble();
11        System.out.println(x:"_____");
12        System.out.println("Sum of two numbers is: " + (firstNum + secondNum));
13        System.out.println("Difference of two numbers is: " + Math.abs(firstNum - secondNum));
14        System.out.println("Product of two numbers is: " + (firstNum * secondNum));
15        System.out.println("Quotient of two numbers is: " + (firstNum / secondNum));
16    }
```

b. Kết quả

```
PS D:\20241\OOP_Lab\IT3103.744530.2024.1.20225849.BuiQuangHung\Lab1> java .\Calculate.java
Bui Quang Hung 20225849 - Please enter first number: 3
Bui Quang Hung 20225849 - Please enter second number: 6

-----
Sum of two numbers is: 9.0
Difference of two numbers is: 3.0
Product of two numbers is: 18.0
Quotient of two numbers is: 0.5
PS D:\20241\OOP_Lab\IT3103.744530.2024.1.20225849.BuiQuangHung\Lab1>
```

2.2.6 Write a program to solve:

a. Code

```
Lab1 > SolveEquation.java > SolveEquation > SystemFirstDegreeEquations()
1 import java.util.Scanner;
2
3 public class SolveEquation {
4     public static void FirstDegreeEquation() {
5         Scanner sc = new Scanner(System.in);
6         double a, b;
7
8         // Input a
9         do {
10            System.out.print("Enter a (a must not be 0): ");
11            a = sc.nextDouble();
12            if (a == 0) {
13                System.out.println("'a' cannot be 0. Please enter a non-zero value.");
14            }
15        } while (a == 0);
16    }
```



```

17 // Input b
18 System.out.print("Enter b: ");
19 while (!sc.hasNextDouble()) {
20     System.out.println("That's not a valid number. Please try again.");
21     sc.next(); // Consume the invalid input
22 }
23 b = sc.nextDouble();
24
25 // Calculate and display the solution
26 double x = -b / a;
27 System.out.println("The solution of the equation is x = " + x);
28 }
29
30 public static void SystemFirstDegreeEquations() {
31     // Enter coefficient
32     // a1*x + b1*y = c1
33     // a2*x + b2*y = c2
34     Scanner sc = new Scanner(System.in);
35     System.out.print("Enter a1: ");
36     double a1 = sc.nextDouble();
37     System.out.print("Enter b1: ");
38     double b1 = sc.nextDouble();
39     System.out.print("Enter c1: ");
40     double c1 = sc.nextDouble();
41     System.out.print("Enter a2: ");
42     double a2 = sc.nextDouble();
43     System.out.print("Enter b2: ");
44     double b2 = sc.nextDouble();
45     System.out.print("Enter c2: ");
46     double c2 = sc.nextDouble();
47
48     // Calculate the determinant coefficient
49     double det = a1 * b2 - a2 * b1;
50
51     // Check if the system of equations has a solution or not
52     if (det == 0) {

```

```

52     if (det == 0) {
53         if (c1 / c2 == a1 / a2) {
54             System.out.println("The system of equations has infinitely many solutions");
55         } else {
56             System.out.println("The system of equations has no solution");
57         }
58     } else {
59         // Calculate x and y
60         double x = (c1 * b2 - c2 * b1) / det;
61         double y = (a1 * c2 - a2 * c1) / det;
62         System.out.println("The solution of the system is (x;y) = (" + x + ", " + y + ")");
63     }
64 }
65
66 public static void SecondDegreeEquations() {
67     Scanner sc = new Scanner(System.in);
68
69     // Enter the coefficients a, b, c
70     double a;
71     do {
72         System.out.print("Enter a (a != 0): ");

```

```

72         System.out.print("Enter a (a != 0): ");
73         a = sc.nextDouble();
74         if (a == 0) {
75             System.out.println("Invalid! Please enter a again:");
76         }
77     } while (a == 0);
78     System.out.print("Enter b: ");
79     double b = sc.nextDouble();
80     System.out.print("Enter c: ");
81     double c = sc.nextDouble();
82
83     // Calculate Delta
84     double delta = b * b - 4 * a * c;
85     if (delta > 0) {
86         // Two real solutions
87         double x1 = (-b + Math.sqrt(delta)) / (2 * a);
88         double x2 = (-b - Math.sqrt(delta)) / (2 * a);
89         System.out.println("The solution of equation is:");
90         System.out.println("x1 = " + x1);
91         System.out.println("x2 = " + x2);
92     } else if (delta == 0) {
93         // Double solution
94         double x = -b / (2 * a);
95         System.out.println("The solution of equation is:");
96         System.out.println("x1 = x2 = " + x);
97     } else {
98         // No solution
99         double re = -b / (2 * a);
100        double im = Math.sqrt(-delta) / (2 * a);
101        System.out.println("The equation has no solution");
102    }
103 }
104
Run | Debug
105 public static void main(String[] args) {
106     Scanner sc = new Scanner(System.in);
107     System.out.println("Hi! I'm Bui Quang Hung 20225849. Please choose one of threes programs: \n"

```

```

107     System.out.println("Hi! I'm Bui Quang Hung 20225849. Please choose one of threes programs: \n"
108     + " 1. Solve the first-degree quation (linear equation) with one variable.\n"
109     + " 2. Solve the system of first-degree quations (linear system) with two variables.\n"
110     + " 3. Solve the second-degree quation with one variable.\n"
111     + " 4. Exit.\n"
112     + "Your chose is: ");
113     int option = sc.nextInt();
114
115     switch (option) {
116     case 1:
117         FirstDegreeEquation();
118         main(null);
119         break;
120     case 2:
121         SystemFirstDegreeEquations();
122         main(null);
123         break;
124     case 3:
125         SecondDegreeEquations();
126         main(null);
127         break;
128     case 4:
129         System.out.println("Program is stopping!");
130         System.exit(option);
131     default:
132         System.out.println("Invalid option. Please choose again.");
133     }
134 }
135 }

```

b. Kết quả

TH1: The first-degree equation (linear equation) with one variable.

```
PS D:\20241\OOP_Lab\IT3103.744530.2024.1.20225849.BuiQuangHung\Lab1> java .\SolveEquation.java
Hi! I'm Bui Quang Hung 20225849. Please choose one of threes programs:
  1. Solve the first-degree quation (linear equation) with one variable.
  2. Solve the system of first-degree quations (linear system) with two variables.
  3. Solve the second-degree quation with one variable.
  4. Exit.
Your chose is:
1
Enter a (a must not be 0): 0
'a' cannot be 0. Please enter a non-zero value.
Enter a (a must not be 0): 1
Enter b: 2
The solution of the equation is  $x = -2.0$ 
```

TH2: The system of first-degree equations (linear system) with two variables.

- Trường hợp hệ phương trình vô nghiệm

```
PS D:\20241\OOP_Lab\IT3103.744530.2024.1.20225849.BuiQuangHung\Lab1> java .\SolveEquation.java
Hi! I'm Bui Quang Hung 20225849. Please choose one of threes programs:
  1. Solve the first-degree quation (linear equation) with one variable.
  2. Solve the system of first-degree quations (linear system) with two variables.
  3. Solve the second-degree quation with one variable.
  4. Exit.
Your chose is:
2
Enter a1: 1
Enter b1: 1
Enter c1: 1
Enter a2: 1
Enter b2: 1
Enter c2: 2
The system of equations has no solution
```

- Trường hợp hệ phương trình vô số nghiệm

```
Hi! I'm Bui Quang Hung 20225849. Please choose one of threes programs:
  1. Solve the first-degree quation (linear equation) with one variable.
  2. Solve the system of first-degree quations (linear system) with two variables.
  3. Solve the second-degree quation with one variable.
  4. Exit.
Your chose is:
2
Enter a1: 1
Enter b1: 1
Enter c1: 1
Enter a2: 3
Enter b2: 3
Enter c2: 3
The system of equations has infinitely many solutions
```

- Trường hợp có nghiệm

```

Hi! I'm Bui Quang Hung 20225849. Please choose one of threes programs:
 1. Solve the first-degree quation (linear equation) with one variable.
 2. Solve the system of first-degree quations (linear system) with two variables.
 3. Solve the second-degree quation with one variable.
 4. Exit.
Your chose is:
2
Enter a1: 1
Enter b1: 3
Enter c1: 5
Enter a2: 2
Enter b2: 4
Enter c2: 7
The solution of the system is (x;y) = (0.5,1.5)

```

TH3: The second-degree equation with one variable

- Trường hợp phương trình có hai nghiệm phân biệt

```

PS D:\20241\OOP_Lab\IT3103.744530.2024.1.20225849.BuiQuangHung\Lab1> java .\SolveEquation.java
Hi! I'm Bui Quang Hung 20225849. Please choose one of threes programs:
 1. Solve the first-degree quation (linear equation) with one variable.
 2. Solve the system of first-degree quations (linear system) with two variables.
 3. Solve the second-degree quation with one variable.
 4. Exit.
Your chose is:
3
Enter a (a != 0): 0
Invalid! Please enter a again:
Enter a (a != 0): 1
Enter b: 5
Enter c: -6
The solution of equation is:
x1 = 1.0
x2 = -6.0

```

- Trường hợp phương trình có nghiệm kép

```

PS D:\20241\OOP_Lab\IT3103.744530.2024.1.20225849.BuiQuangHung\Lab1> java .\SolveEquation.java
Hi! I'm Bui Quang Hung 20225849. Please choose one of threes programs:
 1. Solve the first-degree quation (linear equation) with one variable.
 2. Solve the system of first-degree quations (linear system) with two variables.
 3. Solve the second-degree quation with one variable.
 4. Exit.
Your chose is:
3
Enter a (a != 0): 1
Enter b: -6
Enter c: 9
The solution of equation is:
x1 = x2 = 3.0

```

- Exit Menu

```

Hi! I'm Bui Quang Hung 20225849. Please choose one of threes programs:
1. Solve the first-degree quation (linear equation) with one variable.
2. Solve the system of first-degree quations (linear system) with two variables.
3. Solve the second-degree quation with one variable.
4. Exit.
Your chose is:
4
Program is stopping!
PS D:\20241\OOP_Lab\IT3103.744530.2024.1.20225849.BuiQuangHung\Lab1>

```

B. Exercise

6.1 Write, compile and run the ChoosingOption program

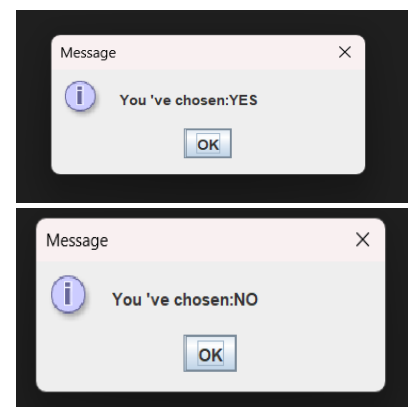
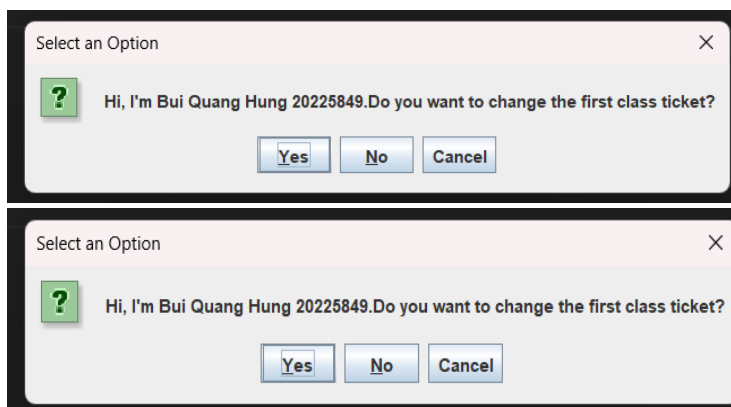
a. Code

```

ChoosingOption.java U X
Lab1 > ChoosingOption.java > ChoosingOption > main(String[])
1
2 import javax.swing.JOptionPane;
3
4 public class ChoosingOption {
5     public static void main(String[] args) {
6         int option = JOptionPane.showConfirmDialog(parentComponent:null,
7             message:"Do you want to change the first class ticket?");
8
9         JOptionPane.showMessageDialog(parentComponent:null,
10            "You 've chosen:" + (option==JOptionPane.YES_OPTION?"YES": "NO"));
11        System.exit(status:0);
12    }
13 }
14

```

b. Kết quả



- Answer questions:

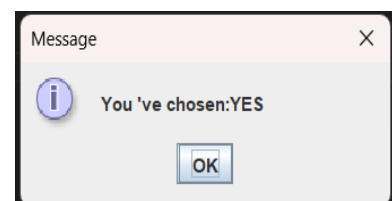
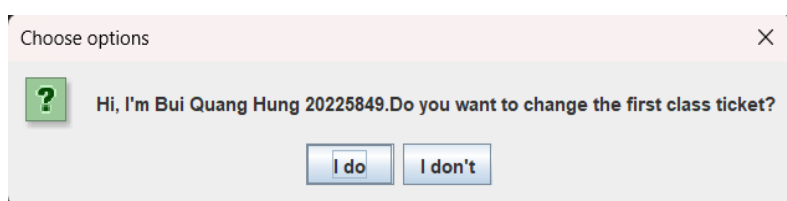
+ What happens if users choose “Cancel”?

- Nếu người dùng chọn "Cancel" trong **JOptionPane.showConfirmDialog**, phương thức **showConfirmDialog** sẽ trả về **JOptionPane.CANCEL_OPTION**, và mã sẽ hiển thị thông báo “You’ve choosen: No” vì điều kiện (`option == JOptionPane.YES_OPTION`) sẽ sai.

+ How to customize the options to users, e.g. only two options: “Yes” and “No”, OR “I do” and “I don’t” ?

- Để tùy chỉnh các tùy chọn cho người dùng, bạn có thể sử dụng phương thức **showOptionDialog** thay vì **showConfirmDialog**. Phương thức **showOptionDialog** cho phép bạn chỉ định các tùy chọn nút tùy chỉnh.

```
ChoosingOption.java U X
Lab1 > ChoosingOption.java > ChoosingOption
1
2 import javax.swing.JOptionPane;
3
4 public class ChoosingOption {
5     public static void main(String[] args) {
6         Object[] options = {"I do", "I don't"};
7         int option = JOptionPane.showOptionDialog(parentComponent:null,
8         message:"Hi, I'm Bui Quang Hung 20225849.Do you want to change the first class ticket?",
9         title:"Choose options", JOptionPane.YES_NO_OPTION, JOptionPane.QUESTION_MESSAGE, icon:null, options,
10        options[0]);
11
12        JOptionPane.showMessageDialog(parentComponent:null,
13        "You 've chosen:" + (option==JOptionPane.YES_OPTION?"YES": "NO"));
14        System.exit(status:0);
15    }
16 }
17
```



6.2 Write a program for input/output from keyboard

a. Code

```
InputFromKeyboard.java 1, U X
Lab1 > InputFromKeyboard.java > ...
1  import java.util.Scanner;
2  public class InputFromKeyboard{
    Run | Debug
3      public static void main(String[] args) {
4          Scanner keyboard = new Scanner(System.in);
5          System.out.println(x:"Hi, I'm Bui Quang Hung. What's your name?");
6          String strName = keyboard.nextLine();
7          System.out.println(x:"How old are you?");
8          int iAge = keyboard.nextInt();
9          System.out.println(x:"How tall are you?");
10         double dHeight = keyboard.nextDouble();
11         System.out.println("Mr/Mrs. " + strName + ", " + iAge + " years old. "
12         + "Your height is " + dHeight + ".");
13     }
14 }
15
16
```

b. Kết quả:

```
PS D:\20241\OOP_Lab\IT3103.744530.2024.1.20225849.BuiQuangHung\Lab1> java .\InputFromKeyboard.java
Hi, I'm Bui Quang Hung. What's your name?
Bui Quang Hung
How old are you?
20
How tall are you?
174
Mr/Mrs. Bui Quang Hung, 20 years old. Your height is 174.0.
PS D:\20241\OOP_Lab\IT3103.744530.2024.1.20225849.BuiQuangHung\Lab1>
```

6.3 Write a program to display a triangle with a height of n stars (*), n is entered by users

a. Code

```
Stars.java 1, U X
Lab1 > Stars.java > ...
1
2  import java.util.Scanner;
3  public class Stars {
    Run | Debug
4      public static void main(String[] args) {
5          Scanner keyboard = new Scanner(System.in);
6          System.out.println(x:"20225849-Enter the n");
7          int n = keyboard.nextInt();
8          for (int i = 1; i <= n; i++) {
9              for(int j = 1 ; j <= n-i ; j++){
10                 System.out.print(s:" ");
11             }
12             for(int k=1 ; k <= 2*i - 1 ; k++){
13                 System.out.print(s:"*");
14             }
15             System.out.println(x:"");
16         }
17     }
18
19 }
20
```

b. Kết quả:

```
PS D:\20241\OOP_Lab\IT3103.744530.2024.1.20225849.BuiQuangHung\Lab1> java .\Stars.java
20225849-Enter the n
5
*
***
*****
*****
*****
PS D:\20241\OOP_Lab\IT3103.744530.2024.1.20225849.BuiQuangHung\Lab1>
```

```
PS D:\20241\OOP_Lab\IT3103.744530.2024.1.20225849.BuiQuangHung\Lab1> java .\Stars.java
20225849-Enter the n
8
*
***
*****
*****
*****
*****
*****
*****
PS D:\20241\OOP_Lab\IT3103.744530.2024.1.20225849.BuiQuangHung\Lab1>
```

6.4 Write a program to display the number of days of a month

a. Code

```
1  import java.util.Scanner;
2
3  public class DaysInMonth {
4      // Check if the month input is valid
5      private static boolean isValidMonth(String month) {
6          return month.matches("(?i)^(january|jan\\.?.?|1|"
7              + "february|feb\\.?.?|2|"
8              + "march|mar\\.?.?|3|"
9              + "april|apr\\.?.?|4|"
10             + "may|5|"
11             + "june|jun\\.?.?|6|"
12             + "july|jul\\.?.?|7|"
13             + "august|aug\\.?.?|8|"
14             + "september|sept?\\.?.?|9|"
15             + "october|oct\\.?.?|10|"
16             + "november|nov\\.?.?|11|"
17             + "december|dec\\.?.?|12)$");
18      }
19
20      // Check if the year input is valid
21      private static boolean isValidYear(String year) {
22          return year.matches("^\\d{4}$");
23      }
24
25      // Check the leap year
26      private static boolean isLeapYear(int year) {
27          return (year % 4 == 0 && year % 100 != 0) || (year % 400 == 0);
28      }
29
30      // Function to get month number
31      private static int getMonthNumber(String monthStr) {
32          monthStr = monthStr.toLowerCase().replaceAll("\\.$", ""); // Remove trailing period if present
33          String[] monthNames = {"january", "february", "march", "april", "may", "june", "july", "august",
34              "september", "october", "november", "december"};
35          String[] monthAbbreviations = {"jan", "feb", "mar", "apr", "may",
36              "jun", "jul", "aug", "sep", "oct", "nov", "dec"};
37
38          for (int i = 0; i < monthNames.length; i++) {
39              if (monthStr.equals(monthNames[i]) || monthStr.equals(monthAbbreviations[i]) ||
```



```

38     for (int i = 0; i < monthNames.length; i++) {
39         if (monthStr.equals(monthNames[i]) || monthStr.equals(monthAbbreviations[i]) ||
40             monthStr.equals(String.valueOf(i + 1))) {
41             return i + 1;
42         }
43     }
44     return -1;
45 }
46
47 // Function to get days in a month
48 private static int getDaysInMonth(int month, int year) {
49     int[] daysInMonth = {31, 28, 31, 30, 31, 30, 31, 31, 30, 31, 30, 31};
50
51     if (month == 2 && isLeapYear(year)) {
52         return 29;
53     }
54
55     return daysInMonth[month - 1];
56 }
57
58 Run | Debug
59 public static void main(String[] args) {
60     Scanner sc = new Scanner(System.in);
61     String monthInput;
62     int year;
63
64     // Input month
65     while (true) {
66         System.out.println("Enter the month (name, abbreviation, or number:");
67         monthInput = sc.nextLine().trim();
68         if (isValidMonth(monthInput)) {
69             break;
70         } else {
71             System.out.println("Invalid month input. Please try again.");
72         }
73     }

```

```

73
74     // Input year
75     while (true) {
76         System.out.println("Enter the year (4 digits:");
77         String yearInput = sc.nextLine().trim();
78         if (isValidYear(yearInput)) {
79             year = Integer.parseInt(yearInput);
80             break;
81         } else {
82             System.out.println("Invalid year input. Please try again.");
83         }
84     }
85
86     int month = getMonthNumber(monthInput);
87     int days = getDaysInMonth(month, year);
88     System.out.printf("Number of days in %s %d: %d%n", monthInput, year, days);
89
90     sc.close();
91 }
92 }

```

b. Kết quả

- Trường hợp input là số

```

● PS D:\20241\OOP_Lab\IT3103.744530.2024.1.20225849.BuiQuangHung\Lab1> java .\DaysInMonth.java
Enter the month (name, abbreviation, or number):
hello
Invalid month input. Please try again.
Enter the month (name, abbreviation, or number):
2
Enter the year (4 digits):
2024
Number of days in 2 2024: 29
○ PS D:\20241\OOP_Lab\IT3103.744530.2024.1.20225849.BuiQuangHung\Lab1>

```

- Trường hợp input là tên đầy đủ của tháng

```

● PS D:\20241\OOP_Lab\IT3103.744530.2024.1.20225849.BuiQuangHung\Lab1> java .\DaysInMonth.java
Enter the month (name, abbreviation, or number):
February
Enter the year (4 digits):
1a1a
Invalid year input. Please try again.
Enter the year (4 digits):
aaa
Invalid year input. Please try again.
Enter the year (4 digits):
24
Invalid year input. Please try again.
Enter the year (4 digits):
2024
Number of days in February 2024: 29
○ PS D:\20241\OOP_Lab\IT3103.744530.2024.1.20225849.BuiQuangHung\Lab1>

```

```

● PS D:\20241\OOP_Lab\IT3103.744530.2024.1.20225849.BuiQuangHung\Lab1> java .\DaysInMonth.java
Enter the month (name, abbreviation, or number):
February
Enter the year (4 digits):
2023
Number of days in February 2023: 28
○ PS D:\20241\OOP_Lab\IT3103.744530.2024.1.20225849.BuiQuangHung\Lab1>

```

- Trường hợp input là tên viết tắt của tháng

```

● PS D:\20241\OOP_Lab\IT3103.744530.2024.1.20225849.BuiQuangHung\Lab1> java .\DaysInMonth.java
Enter the month (name, abbreviation, or number):
Jan.
Enter the year (4 digits):
1999
Number of days in Jan. 1999: 31
○ PS D:\20241\OOP_Lab\IT3103.744530.2024.1.20225849.BuiQuangHung\Lab1>

```

```

● PS D:\20241\OOP_Lab\IT3103.744530.2024.1.20225849.BuiQuangHung\Lab1> java .\DaysInMonth.java
Enter the month (name, abbreviation, or number):
jan
Enter the year (4 digits):
1999
Number of days in jan 1999: 31
○ PS D:\20241\OOP_Lab\IT3103.744530.2024.1.20225849.BuiQuangHung\Lab1>

```

6.5 Write a Java program to sort a numeric array, and calculate the sum and average value of array elements.

a. Code

```
Lab1 > ArrayOfNumbers.java > ArrayOfNumbers > main(String[])
1  import java.util.Scanner;
2
3  public class ArrayOfNumbers {
4      // Function to sort elements in an array
5      public static void bubbleSort (int[] arr) {
6          int n = arr.length;
7          for (int i = 0 ; i < n - 1 ; i++) {
8              for(int j = 0 ; j < n - i - 1 ; j++){
9                  if (arr[j] > arr[j+1]) {
10                     int temp = arr[j];
11                     arr[j] = arr[j + 1];
12                     arr[j + 1] = temp;
13                 }
14             }
15         }
16     }
17     // Function to sum elements in array
18     public static int calculateSum(int[] arr) {
19         int sum = 0;
20         for (int num : arr) {
21             sum += num;
22         }
23         return sum;
24     }
25     // Function to calculate the average value of elements in an array
26     public static double calculateAverage(int[] arr) {
27         int sum = calculateSum(arr);
28         return (double) sum / arr.length;
29     }
30     // Function to print the elements of array
31     public static void printArray (int[] arr, int n) {
32         for (int i = 0 ; i < n ; i++){
33             System.out.printf("%d ", arr[i]);
34         }
35         System.out.println("");
36     }
37     Run | Debug
38     public static void main(String[] args) {
39         Scanner sc = new Scanner(System.in);
```

```

37 public static void main(String[] args) {
38     Scanner sc = new Scanner(System.in);
39
40     // Enter the size of array
41     System.out.println("Enter the size of array");
42     int n = sc.nextInt();
43
44     // Array declaration
45     int array[] = new int[n];
46     System.out.println("Enter the elements of array:");
47     for (int i = 0; i < n; i++) {
48         System.out.print("Enter the element " + (i + 1) + ": ");
49         array[i] = sc.nextInt();
50     }
51
52     System.out.println("Inputed Array: ");
53     printArray(array, n);
54     System.out.println("Sorted Array:");
55     bubbleSort(array);
56     printArray(array, n);
57     System.out.println("The sum of array is " + calculateSum(array));
58     System.out.println("The average of array is " + calculateAverage(array));
59
60     sc.close();
61 }
62 }

```

b. Kết quả

```

PS D:\20241\OOP_Lab\IT3103.744530.2024.1.20225849.BuiQuangHung\Lab1> java .\ArrayOfNumbers.java
Enter the size of array
5
Enter the elements of array:
Enter the element 1: 3
Enter the element 2: 5
Enter the element 3: 1
Enter the element 4: 7
Enter the element 5: 9
Inputed Array:
3 5 1 7 9
Sorted Array:
1 3 5 7 9
The sum of array is 25
The average of array is 5.0

```

```

PS D:\20241\OOP_Lab\IT3103.744530.2024.1.20225849.BuiQuangHung\Lab1> java .\ArrayOfNumbers.java
Enter the size of array
3
Enter the elements of array:
Enter the element 1: 2
Enter the element 2: 1
Enter the element 3: 4
Inputed Array:
2 1 4
Sorted Array:
1 2 4
The sum of array is 7
The average of array is 2.3333333333333335
PS D:\20241\OOP_Lab\IT3103.744530.2024.1.20225849.BuiQuangHung\Lab1>

```

6.6 Write a Java program to add two matrices of the same size

a. Code

```

Lab1 > AddTwoMatrices.java > ...
1  import java.util.Scanner;
2
3  public class AddTwoMatrices {
4      public static void main(String[] args) {
5          Scanner sc = new Scanner(System.in);
6
7          System.out.println("Enter the number of rows for the matrices ");
8          int rows = sc.nextInt();
9          System.out.println("Enter the number of cols for the matrices ");
10         int cols = sc.nextInt();
11
12         int[][] firstMatrix = new int[rows][cols];
13         int[][] secondMatrix = new int[rows][cols];
14         int[][] sumMatrix = new int[rows][cols];
15
16         System.out.println("Enter the elements of the first matrix:");
17         enterMatrix(firstMatrix, rows, cols);
18
19         System.out.println("Enter the elements of the second matrix:");
20         enterMatrix(secondMatrix, rows, cols);
21         //add two matrices
22         for(int i = 0 ; i < rows ; i++) {
23             for(int j = 0 ; j < cols ; j++) {
24                 sumMatrix[i][j] = firstMatrix[i][j] + secondMatrix[i][j];
25             }
26         }
27
28         System.out.println("The result of matrix addition is:");
29         printMatrix(sumMatrix, rows, cols);
30
31         sc.close();
32     }

```

```

33     public static void enterMatrix (int[][] matrix, int rows, int cols) {
34         Scanner sc = new Scanner(System.in);
35         for(int i = 0 ; i < rows ; i++) {
36             for(int j = 0 ; j < cols ; j++) {
37                 matrix[i][j] = sc.nextInt();
38             }
39         }
40     }
41     public static void printMatrix (int[][] matrix, int rows, int cols) {
42         for(int i = 0 ; i < rows ; i++) {
43             for(int j = 0 ; j < cols ; j++) {
44                 System.out.print(matrix[i][j] + " ");
45             }
46             System.out.println();
47         }
48     }
49 }

```

b. Kết quả

```

PS D:\20241\OOP_Lab\IT3103.744530.2024.1.20225849.BuiQuangHung\Lab1> java .\AddTwoMatrices.java
Enter the number of rows for the matrices
2
Enter the number of cols for the matrices
3
Enter the elements of the first matrix:
1 3 5
7 9 11
Enter the elements of the second matrix:
2 4 6
8 10 12
The result of matrix addition is:
3 7 11
15 19 23

```